Venous thromboembolism (DVT / PE)

Diagnose with pretest probability → D-dimer → imaging · triage PE by ESC risk class · **3 months for everyone, then stop for a major transient provocation or extend (often reduced-dose DOAC) for unprovoked or persistent risk** · DOACs first-line; LMWH or warfarin in pregnancy, APS and selected cancers

Compiled from the Fellowship DVT/PE, Anticoagulation & Reversal and Hematology in Pregnancy reviews · ESC 2019 PE, ASH 2018–2021, ASCO 2023, ISTH SSC guidance · FDA labels · primary trial reports verified against published abstracts, October 2026.

DIAGNOSIS

- **Pretest probability first.** Wells DVT **≤1 → D-dimer** (negative excludes), **≥2 → compression ultrasound**. Wells PE **≤4 → D-dimer**, **>4 → CTPA**
- **PERC** only if gestalt <15%: age <50, HR <100, SaO_2 ≥95%, no hemoptysis, estrogen, prior VTE, leg swelling, recent surgery/trauma → no testing
- **Age-adjusted D-dimer: age × 10 µg/L** if >50 y *ADJUST-PE*. **YEARS:** no items (DVT signs, hemoptysis, PE most likely) → exclude if <1000 ng/mL; ≥1 item → <500; CTPA avoided in 48%
- **D-dimer rules out, never in** — skip it when probability is high
- **Imaging:** CUS (repeat in 1 wk if negative proximal scan + high suspicion). **CTPA** for PE; **V/Q** if contrast allergy, renal failure, or pregnancy with normal CXR. Unstable: bedside echo + CUS

ACUTE PE RISK — ESC 2019

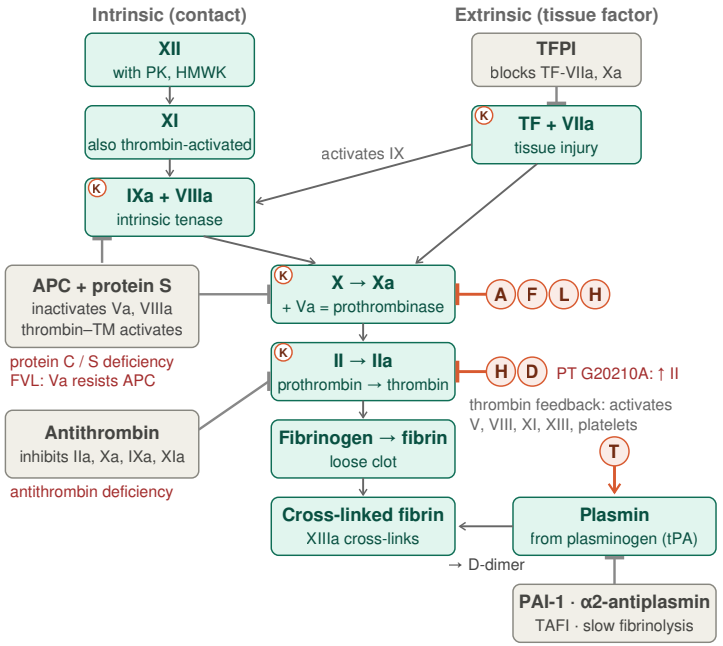
Class	Defined by	Management
High	**Hemodynamic instability**: arrest, SBP <90 or pressors, or SBP drop ≥40 for >15 min	UFH + **systemic lysis**; embolectomy (surgical/catheter) if contraindicated or fails
Intermediate-high	sPESI ≥1 + **RV dysfunction AND raised troponin**	Anticoagulate, monitor; **rescue lysis or catheter therapy if decompensates**; PERT
Intermediate-low	sPESI ≥1 + one or neither, or sPESI 0 + either positive	Admit, anticoagulate
Low	**sPESI 0**, no RV dysfunction/troponin	**Home or early discharge** if Hestia-negative

sPESI (1 each): age >80, cancer, chronic heart/lung disease, HR ≥110, SBP <100, SaO_2 <90%. **RV dysfunction:** RV/LV ≥1.0 on CT or echo dilation/hypokinesis. **Hestia:** 11 exclusions (O_2 need, bleeding risk, CrCl <30, pregnancy, social…).

ACUTE PE — THERAPIES

- **Systemic lysis for high-risk PE only:** alteplase **100 mg over 2 h**. Not routine in intermediate-high: tenecteplase cut decompensation 2.6% vs 5.6% but **hemorrhagic stroke 2.0% vs 0.2%** *PEITHO*
- **Catheter-directed therapy:** lysis contraindicated, or intermediate-high deteriorating — PERT decision. **UFH** first if lysis/procedure likely
- **IVC filter only if acute proximal DVT/PE + anticoagulation contraindicated**; retrievable — anticoagulate and remove once safe. Not for recurrence on therapy or added to anticoagulation

PATHOPHYSIOLOGY



(A) apixaban · rivaroxaban · edoxaban: direct Xa
(F) fondaparinux: Xa via AT
(L) LMWH: mostly Xa via AT
(H) UFH: IIa + Xa via AT
(D) dabigatran · argatroban · bivalirudin: direct IIa
(T) alteplase: activates plasmin
(K) warfarin: blocks vitamin K-dependent II, VII, IX, X (+ proteins C, S)

gray = natural anticoagulants · red = thrombophilia defects

Thrombophilias weaken the brakes on Va, VIIIa and thrombin. Warfarin lowers protein C before II/X: overlap with a parenteral agent ≥5 d and until INR ≥2.

PROVOKED VS UNPROVOKED — ISTH 2016 / ESC 2019

Category	Examples	Default
Major transient *≤3 mo*	Surgery >30 min general anesthesia; bed-bound admission ≥3 d; cesarean; major trauma	**3 mo, stop**
Minor transient *≤2 mo*	Short surgery or admission; **estrogen**, pregnancy; leg injury; long-haul travel	**3 mo**, usually stop
Persistent	**Active cancer**, IBD, autoimmune disease, APS	**Extended**
Unprovoked	No identifiable factor	**Extended** if bleeding risk OK

Recurrence after stopping, first unprovoked VTE: 10% at 1 y, **25% at 5 y, 36% at 10 y**; higher in men *Khan BMJ 2019*. Age-appropriate cancer screening only.

TREATMENT PHASES AND DURATION

Initiation + treatment phase

Start full dose
Apixaban 10 mg BID ×7 d → 5 mg BID
Rivaroxaban 15 mg BID ×21 d → 20 mg daily
Edoxaban/dabigatran: 5–10 d LMWH first
Warfarin: overlap ≥5 d + INR ≥2 ×24 h

Treat 3 months for everyone
(cancer: ≥6 months) — then decide

Decision at 3 months

Transient factor
major → **stop**; minor → usually stop, extend only if the risk persists

Unprovoked or persistent factor
extended, no stop date if bleeding risk low–moderate; **reassess every year**
2nd unprovoked VTE → indefinite

Extended phase (after ≥6 months)

Reduced-dose DOAC
apixaban 2.5 mg BID (AMPLIFY-EXT)
rivaroxaban 10 mg daily (EINSTEIN-CHOICE)
cancer: apixaban 2.5 BID noninferior (API-CAT)

Full dose or aspirin
full dose if recurrence on reduced dose (RENOVE: reduced dose missed NI, but less bleeding)
aspirin only if stopping: ~32% fewer recurrences

EXTENDED-PHASE EVIDENCE

- **Apixaban 2.5 or 5 mg BID vs placebo:** recurrence **1.7% / 1.7% vs 8.8%**, no increase in major bleeding *AMPLIFY-EXT*
- **Rivaroxaban 10 or 20 mg vs aspirin 100 mg: 1.2% / 1.5% vs 4.4%** *EINSTEIN-CHOICE*
- **Higher recurrence risk:** male sex, proximal DVT or PE (vs distal), D-dimer high after stopping, obesity, APS, active cancer

Venous thromboembolism (DVT / PE)

CHOOSING THE ANTICOAGULANT

Agent	Choose when	Avoid / caution
Apixaban	**Default**; no lead-in; ~27% renal — best DOAC in CKD and GI cancer; major bleeding RR 0.31 vs LMWH/warfarin (AMPLIFY)	APS; strong dual inducers
Rivaroxaban	No lead-in; daily after day 21	**With food**; CrCl <15; heavy menses
Edoxaban	Daily, after parenteral lead-in	GI cancer bleeding
Dabigatran	After lead-in; idarucizumab reversal	**CrCl ≤30**; dyspepsia
Warfarin	**APS**, mechanical valve, dialysis, cost	**Pregnancy**; INR checks
LMWH	**Pregnancy**; selected cancers; **recurrence on DOAC**	CrCl <30 (adjust/UFH)
Fondaparinux	HIT history (off-label); SVT	CrCl <30; no reversal
UFH	Lysis likely; CrCl <30; high bleeding risk	HIT; monitoring

CANCER-ASSOCIATED VTE — ASCO 2023, ASH 2021

- **DOAC or LMWH ≥6 mo, then while cancer is active.** Treat incidental clot like symptomatic
- **DOACs cut recurrence vs dalteparin; edoxaban/rivaroxaban add GI/GU bleeding:** edoxaban major bleeding **6.9% vs 4.0%** (Hokusai); rivaroxaban recurrence 4% vs 11%, CRNMB 13% vs 4% (SELECT-D). **Apixaban no excess**: 3.8% vs 4.0% (Caravaggio), 0% vs 1.4% (ADAM VTE)
- **LMWH (or apixaban)** for unresected luminal GI/GU tumors, platelets <50k, poor intake, interacting drugs (enzalutamide, apalutamide, azoles, ritonavir)
- **Extended:** apixaban **2.5 mg BID** after ≥6 mo — recurrence 2.1% vs 2.8%, bleeding 12.1% vs 15.6% *API-CAT*
- **Prophylaxis:** starting chemo + **Khorana ≥2** + low bleeding risk → **apixaban 2.5 BID or rivaroxaban 10 daily**. AVERT 4.2% vs 10.2%; CASSINI NS at 180 d (6.0% vs 8.8%). **Myeloma + IMiD:** aspirin or LMWH/DOAC by risk

KHORANA SCORE

Factor	Points
Very high-risk site: stomach, pancreas	2
High-risk site: lung, lymphoma, gynecologic, bladder, testicular	1
Pre-chemo platelets ≥350k · Hb <10 or ESA · WBC >11k · BMI ≥35	1 each

0 low, 1–2 intermediate, ≥3 high. Trials and ASCO 2023 use **≥2** as the prophylaxis threshold.

SPECIAL POPULATIONS

Setting	Approach
APS	**Warfarin INR 2–3; no DOACs**, above all triple-positive or arterial events. TRAPS (rivaroxaban, triple-positive): thromboembolism **12% vs 0**, stopped early; ASTRO-APS (apixaban): excess strokes
Pregnancy	**Therapeutic LMWH by weight**; no routine anti-Xa. **No warfarin or DOAC antepartum.** Treat to **≥6 wk postpartum, ≥3 mo total**. Planned delivery: last therapeutic dose **24 h** before neuraxial (prophylactic 12 h). Postpartum LMWH or warfarin (breastfeeding OK); DOAC only if not breastfeeding
Prior VTE, now pregnant	Antepartum prophylaxis if prior VTE was unprovoked or hormonal; **postpartum prophylaxis 6 wk** for most. **Fixed low-dose LMWH** is enough — weight-adjusted intermediate dose no better (Highlow: 2% vs 3%)
Obesity	**Standard-dose apixaban or rivaroxaban at any BMI/weight** (ISTH 2021); avoid dabigatran/edoxaban at extremes; no routine DOAC levels. **After bariatric surgery:** parenteral first, not a DOAC
CKD	CrCl <30: **apixaban** (limited data), UFH, or enoxaparin 1 mg/kg daily; avoid dabigatran (≤30) and rivaroxaban (<15); edoxaban 30 mg for CrCl 15–50. Dialysis: apixaban or warfarin
Thrombocytopenia	**>50k: full dose.** 25–50k: acute, high-risk clot → full-dose LMWH + platelet transfusion to 40–50k; otherwise 50% or prophylactic LMWH. **<25k: hold** *ISTH 2018*. Avoid DOACs <50k
HIT	Stop all heparin (flushes too); **argatroban, bivalirudin, fondaparinux or a DOAC**. **No warfarin until platelets recover** (≥150k), then overlap ≥5 d; vitamin K if on warfarin at diagnosis. Bilateral CUS. With thrombosis treat 3 mo
Elderly / frail	DOAC over warfarin; apixaban lowest bleeding. **AF dose-reduction criteria do not apply to VTE** — apixaban 5 mg BID regardless of age/weight/creatinine
Recurrence on therapy	Confirm it is new clot (compare with baseline imaging). Check **adherence, dose, interactions, absorption**; look for **cancer, APS, HIT, MPN/PNH**, May-Thurner. **DOAC/warfarin → therapeutic LMWH; on LMWH → raise dose 20–25%**

RECURRENCE AND BLEEDING RISK

- **HERDOO2** (women, first unprovoked VTE, after 5–12 mo): post-thrombotic signs, **D-dimer ≥250 µg/L** on treatment, BMI ≥30, age ≥65. **≤1 → may stop**: 3.0%/pt-yr vs 8.1% in high-risk women + men *REVERSE II*
- **Bleeding risk:** age >65–75, prior bleeding, cancer, renal/liver failure, low platelets, prior stroke, antiplatelets, alcohol, anemia → **high risk favors stopping at 3 mo**

UNUSUAL SITES AND DISTAL CLOT

Site	Approach
Splanchnic	Non-cirrhotic: **JAK2 V617F** (then CALR, marrow), **PNH flow**, APS; local causes. Treat ≥3–6 mo; **indefinite if MPN, unprovoked, Budd-Chiari**. Screen varices without delaying
Cerebral venous	**Anticoagulate even with hemorrhagic infarct.** Heparin → **DOAC acceptable** (RE-SPECT CVT, n=120; ACTION-CVT cohort: less major hemorrhage, HR 0.35). 3–12 mo
Arm / catheter	**3 mo or while the line stays**; **keep a working, needed line**. Paget-Schroetter: CDT + thoracic outlet decompression
Superficial vein	**≥5 cm, >3 cm from SFJ → fondaparinux 2.5 mg ×45 d** (CALISTO 0.9% vs 5.9%). **≤3 cm from SFJ → treat as DVT**
Isolated distal DVT	Mild, no risk factors → **serial CUS over 2 wk**. Treat 3 mo if severe symptoms, extensive, cancer, inpatient, prior VTE. CACTUS: nadroparin no better (3% vs 5%), bleeding 4% vs 0%

THROMBOPHILIA TESTING

- **Don't test** if it won't change management: major transient provocation; unprovoked VTE already on extended therapy; cancer; acute clot or on anticoagulation; routine relatives
- **May matter: APS** in young unprovoked VTE, unusual sites, arterial + venous events, pregnancy morbidity — before choosing a DOAC. Women with a relative's VTE + known thrombophilia weighing estrogen/pregnancy
- **Timing:** FVL, PT G20210A any time. **Protein C/S, AT fall with acute clot**; warfarin lowers C/S, heparin AT, pregnancy S; DOACs falsely raise clot-based assays. **LA unreliable on DOAC/warfarin**; ELISAs unaffected. **APS needs persistence ≥12 wk**

THROMBOPHILIAS

Defect	Prevalence · RR first VTE *ASH Ed 2016*	Notes
Factor V Leiden (het)	3–7% of Europeans · **3–5×**	APC resistance; ~20% of VTE pts. Homozygous much higher
Prothrombin G20210A	0.7–4% · **2–3×**	Small effect on recurrence
Antithrombin def.	0.02–0.2% · **5–10×**	Strongest; heparin resistance
Protein C def.	rare · **4–6.5×**	Warfarin skin necrosis — overlap heparin
Protein S def.	rare · **1–10×**	Free S low in pregnancy, estrogen
Acquired	—	**APS**, cancer, MPN (JAK2), PNH, nephrotic syndrome, IBD, HIT, estrogen, pregnancy

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AGENTS — DOSING

Agent	Treatment	Extended · renal
Apixaban	**10 mg BID ×7 d → 5 mg BID**	**2.5 mg BID** after ≥6 mo. No renal or age reduction for VTE
Rivaroxaban	**15 mg BID ×21 d → 20 mg daily**, with food	**10 mg daily** after ≥6 mo. Avoid CrCl <15
Edoxaban	LMWH/UFH ≥5 d → **60 mg daily**	**30 mg** if CrCl 15–50, ≤60 kg, or P-gp inhibitor
Dabigatran	LMWH/UFH 5–10 d → **150 mg BID**	Avoid CrCl ≤30
Warfarin	Start with parenteral; **overlap ≥5 d and until INR ≥2 for 24 h**; target 2–3	Not renally cleared
Enoxaparin	**1 mg/kg q12h** (1.5 mg/kg daily option; avoid in obesity/cancer)	CrCl <30: **1 mg/kg daily**. Peak anti-Xa 0.6–1.0 (q12h) if obese, pregnant, renal
Dalteparin	Cancer: **200 IU/kg daily ×1 mo → 150 IU/kg daily** (max 18,000 IU)	Anti-Xa if CrCl <30
Fondaparinux	**<50 kg 5 mg · 50–100 kg 7.5 mg · >100 kg 10 mg** daily	SVT 2.5 mg ×45 d. Avoid CrCl <30
UFH	**80 U/kg bolus → 18 U/kg/h**; anti-Xa 0.3–0.7	Preferred for CrCl <30 or likely lysis
Argatroban *HIT*	**2 µg/kg/min** (0.5–1.2 if hepatic dysfunction, ICU); aPTT 1.5–3× baseline	Hepatic clearance; raises INR
Alteplase *high-risk PE*	**100 mg IV over 2 h**	Hold UFH during infusion (US label); ESC allows continuing

SWITCHING AGENTS

From → to	How
LMWH → DOAC	Give DOAC when the next LMWH dose is due
UFH infusion → DOAC	Start DOAC when the infusion stops
DOAC → LMWH/UFH	Start when the next DOAC dose is due
Warfarin → DOAC	Stop warfarin; start apixaban/dabigatran at INR <2, edoxaban ≤2.5, rivaroxaban <3
DOAC → warfarin	Overlap until INR ≥2 (edoxaban: halve dose during overlap); DOACs raise the INR — draw at DOAC trough

FOLLOW-UP AFTER VTE

- **Post-thrombotic syndrome** (pain, swelling, skin change, ulcer; score by Villalta): **stockings do not prevent it** — 14.2% vs 12.7% with placebo stockings *SOX 2014*; use them for symptoms. **No routine catheter lysis for proximal DVT**: PTS 47% vs 48%, major bleeding 1.7% vs 0.3% *ATTRACT 2017*
- **CTEPH:** persistent dyspnea or exercise limitation at **3–6 mo after PE** → echo + **V/Q scan** (more sensitive than CTPA for chronic clot) → CTEPH center. No routine screening of asymptomatic patients
- **At 3 months:** recheck bleeding risk, renal function, CBC, adherence; record a baseline CUS/imaging if it would help judge a future recurrence

PERIPROCEDURAL MANAGEMENT

- **DOACs: hold by bleeding risk, no bridging, no pre-op levels** *PAUSE 2019*. Xa inhibitors: **low/moderate risk skip 1 day** before, resume day 1 after; **high risk skip 2 days**, resume day 2–3. Dabigatran with CrCl <50: 2 d / 4 d. Major bleeding ≤2% (~3% high-risk procedures), arterial events <1% (AF population)
- **Warfarin:** stop 5 d before; **no bridging for most** — BRIDGE (AF): arterial events 0.4% vs 0.3%, major bleeding **1.3% vs 3.2%** with LMWH bridging
- **VTE <3 months ago:** defer elective surgery; if it can't wait, bridge with LMWH; if <1 mo and surgery is mandatory, a retrievable filter can be considered
- **Neuraxial:** follow ASRA intervals; prophylactic LMWH 12 h, therapeutic 24 h before the needle

DRUG INTERACTIONS

- **Avoid DOACs with strong combined CYP3A4 + P-gp inducers:** rifampin, carbamazepine, phenytoin, St John's wort, enzalutamide, apalutamide → use LMWH
- **Strong combined inhibitors** (ketoconazole, itraconazole, ritonavir incl. nirmatrelvir/ritonavir, cobicistat): avoid rivaroxaban; apixaban halve the dose (avoid if already on 2.5 mg). **Dabigatran/edoxaban:** P-gp inhibitors (dronedarone, ketoconazole) raise levels
- **Bleeding add-ons:** antiplatelets, NSAIDs, SSRIs, ibrutinib; stop aspirin unless there is a firm arterial indication

HORMONAL THERAPY — VTE RISK BY TYPE

Type	VTE risk
Combined pill	↑ **~3×** (OR 2.97): levonorgestrel / norethisterone / norgestimate ~2.5×; desogestrel, gestodene, drospirenone, cyproterone ~3.6–4.3× *Vinogradova BMJ 2015*
Combined patch / ring	↑ **~8×** patch, ~6.5× ring vs non-users; ~2× vs levonorgestrel pill *Lidegaard BMJ 2012*
Oral HRT	↑ **~1.6×** (hepatic first pass); CEE + MPA 2.1×, estradiol + dydrogesterone 1.2× (NS)
Transdermal estradiol *patch / gel*	**No** ↑ (OR 0.93 *Vinogradova BMJ 2019*; 0.9 *ESTHER*) — preferred; pair with micronized progesterone or dydrogesterone
Progestin-only	LNG-IUD, pill: no ↑; implant ~1.4× (NS); **DMPA ↑ ~2.7×**
Tamoxifen / raloxifene	↑ **~2–3×**; raloxifene ~30% fewer events than tamoxifen (STAR)

Low-dose vaginal estrogen: no clear ↑. Estrogen-containing or progestin-only therapy may continue during therapeutic anticoagulation (no excess recurrence, EINSTEIN — Martinelli Blood 2016); stop estrogen before anticoagulation ends (German guidance: ≥6 wk before). Gender-affirming: estradiol not ethinyl estradiol; transdermal preferred.

KEY TRIALS

Trial	Result → practice
AMPLIFY-EXT *NEJM 2013*	Apixaban 2.5 or 5 mg BID vs placebo, 12 mo after 6–12 mo Rx: recurrent VTE/VTE death **1.7%** / **1.7% vs 8.8%**; major bleeding 0.2 / 0.1 vs 0.5%
EINSTEIN-CHOICE *NEJM 2017*	Rivaroxaban 20 / 10 mg vs aspirin 100 mg (n=3396): recurrence **1.5%** / **1.2% vs 4.4%**; no significant bleeding increase
RENOVE *Lancet 2025*	High-risk, reduced vs full-dose DOAC (n=2768): recurrence 2.2% vs 1.8%, **noninferiority not met**; clinically relevant bleeding **9.9% vs 15.2%**
Hokusai VTE Cancer *NEJM 2018*	Edoxaban vs dalteparin: recurrence/major bleeding 12.8% vs 13.5% (NI); **major bleeding 6.9% vs 4.0%**, mostly GI cancer
SELECT-D *JCO 2018*	Rivaroxaban vs dalteparin (n=406): 6-mo recurrence **4% vs 11%**; CRNMB 13% vs 4%
Caravaggio *NEJM 2020*	Apixaban vs dalteparin (n=1155): recurrence 5.6% vs 7.9% (NI); **major bleeding 3.8% vs 4.0%**
API-CAT *NEJM 2025*	Cancer, after ≥6 mo: apixaban 2.5 vs 5 mg BID (n=1766): recurrence 2.1% vs 2.8% (NI); CR bleeding **12.1% vs 15.6%**
ADAM VTE *JTH 2020*	Apixaban vs dalteparin (n=300): major bleeding **0% vs 1.4%** (NS); recurrence 0.7% vs 6.3%
AVERT *NEJM 2019*	Khorana ≥2, apixaban 2.5 BID vs placebo: VTE **4.2% vs 10.2%**; major bleeding 3.5% on apixaban
CASSINI *NEJM 2019*	Rivaroxaban 10 mg vs placebo: 6.0% vs 8.8% at 180 d (**NS**); on-treatment 2.6% vs 6.4%
TRAPS *Blood 2018*	Triple-positive APS: rivaroxaban thromboembolism **12% vs 0** on warfarin → stopped early, **no DOACs**
PEITHO *NEJM 2014*	Intermediate-risk PE, tenecteplase: death/decompensation 2.6% vs 5.6%; **hemorrhagic stroke 2.0% vs 0.2%** → no routine lysis
REVERSE II *BMJ 2017*	HERDOO2 ≤1 women stopped: recurrence **3.0%/pt-yr**
YEARS *Lancet 2017*	Variable D-dimer threshold: CTPA avoided in 48% of suspected PE
CALISTO *NEJM 2010*	SVT ≥5 cm, fondaparinux 2.5 mg ×45 d: **0.9% vs 5.9%**
CACTUS *Lancet Haem 2016*	Low-risk calf DVT: nadroparin vs placebo 3% vs 5% (NS); bleeding 4% vs 0%
Highlow *Lancet 2022*	Pregnancy, prior VTE: weight-adjusted intermediate vs fixed low-dose LMWH, VTE 2% vs 3% (NS) → **low dose**
SOX / ATTRACT *2014 / 2017*	Stockings and catheter lysis **do not prevent PTS**; ATTRACT more major bleeding
PAUSE / BRIDGE *2019 / 2015*	Standardized DOAC holds, no bridging: major bleeding ≤2% (~3% high-risk procedures), arterial <1%. Bridging warfarin raised major bleeding 1.3% → 3.2%, no fewer arterial events